

Assignment Task Overview

In this assignment, you will use DAX (Data Analysis Expressions) in Power BI to create calculated columns and measures that generate meaningful business metrics. You will then develop interactive visualizations to analyze sales trends, profitability, and performance indicators within the e-commerce business environment.

The focus of this assignment is on DAX calculations, analytical thinking, and visual storytelling, while ensuring that the underlying data model supports accurate reporting and insights.

DAX Calculations

Calculated Columns

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

FileHomeHelpTable toolsColumn tools

NameCategory Type

FormatText

SummarizationDon't summarize

Sort by column

Data typeText

\$ % ∞ ∅ Auto

Data categoryUncategorized

Data groups

Manage relationships

New column

StructureFormattingPropertiesSortGroupsRelationshipsCalculation

1 Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category	SALES AMOUNT
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average	118932
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average	-595
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average	-32038
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	1413	Below Average	785
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	525	Below Average	0
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	100	Below Average	-125
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	129	Below Average	0
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	320	Below Average	-9440
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	14427	Above Average	0
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	2824	Above Average	31770
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average	0
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average	-12963
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average	-221615

- **Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.**

Revenue = 'Order Details'[Amount] * 'Order Details'[Quantity]

File Home Help Table tools Column tools

Name Category Type Format Text Summarization Don't summarize Data category Uncategorized Sort by column Sort Data groups Groups Manage relationships Relationships New column Calculation

Structure Formatting Properties

1 Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category	SALES AMOUNT
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average	118932
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B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	3175	Above Average	-221615

- **Create a Calculated Column to Categorize Sales: Add a calculated column named**

'Sales Category' in the Order Details table that categorizes each order as 'Above

Average' or 'Below Average' based on the Amount value.

Sales Category = VAR AverageAmount = CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details'))

RETURN

IF(

'Order Details'[Amount] > AverageAmount,

"Above Average",

"Below Average"

Power BI interface showing the DAX formula for the Sales Category measure.

Name: Sales Category
Data type: Text
Format: Text
Summarization: Don't summarize
Data category: Uncategorized

Structure:

```
1 Sales Category =  
2 VAR AverageAmount = CALCULATE(AVERAGE('Order Details'[Amount]), ALL('Order Details'))  
3 RETURN  
4 IF(  
5     'Order Details'[Amount] > AverageAmount,  
6     "Above Average",  
7     "Below Average"  
8 )
```

Table:

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average
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B-25622	534	0	3	Clothing	Saree	Clothing - Saree	1602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	596	Below Average
B-25625	625	240	5	Clothing	Saree	Clothing - Saree	2175	Above Average

Calculated Measures

1. Order Count- Create a measure that calculates the total number of orders in the Order

Details table.

Order Count = COUNTROWS('Order Details')

2. Average Profit in Delhi- Create a measure that calculates the average profit for orders

placed in Delhi.

Average Profit Delhi =

CALCULATE (

AVERAGE ('Order Details'[Profit]),

'List of Orders'[City] = "Delhi"

)

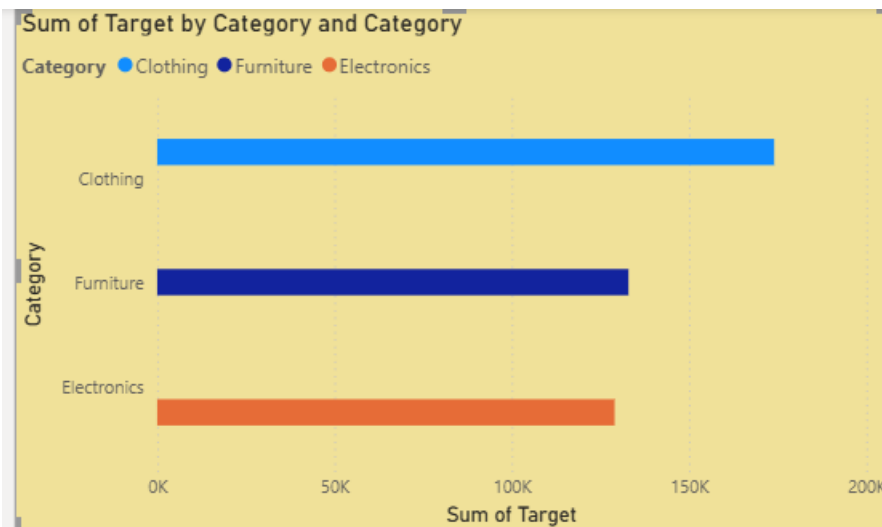
3. Year-to-Date (YTD) Sales- Create a measure that calculates total sales accumulated from the

beginning of the year up to the current order date.

YTD Sales = TOTALYTD(SUM('Order Details'[Amount]), 'List of Orders'[SALES DATE])

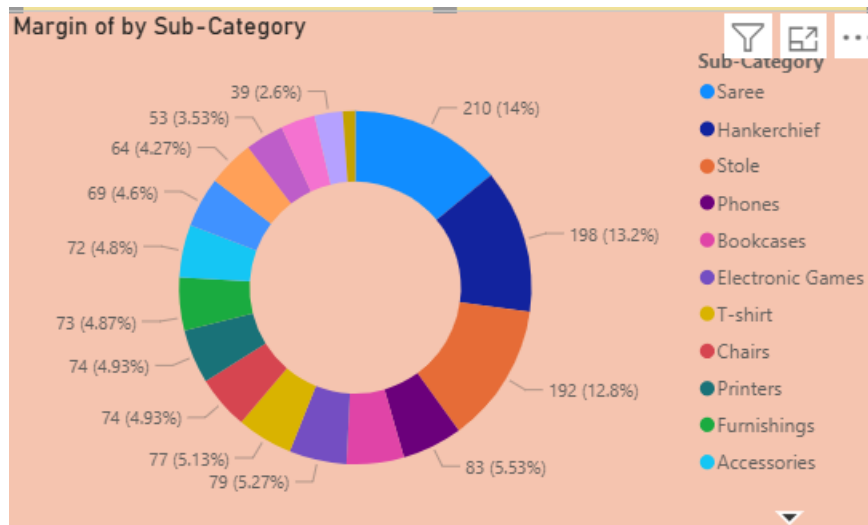
Data Visualization

- **Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.**



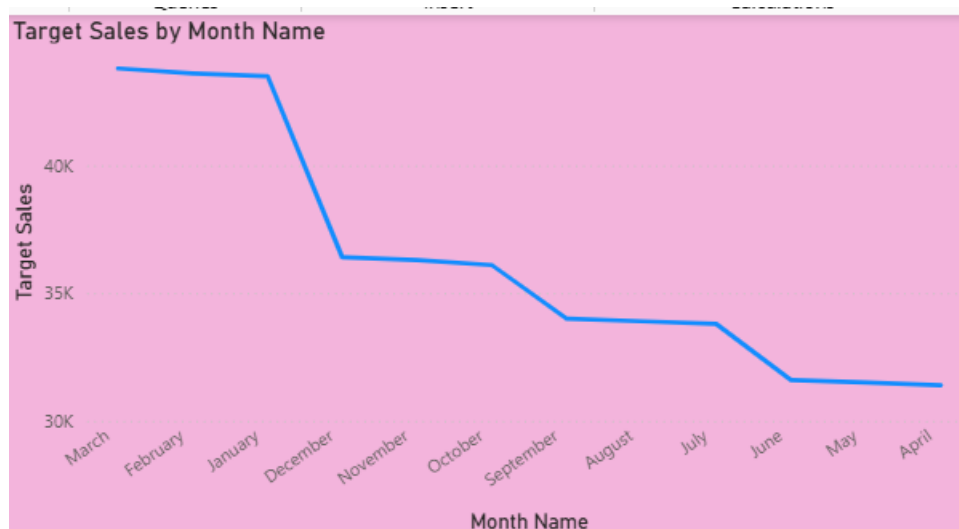
- Technology category achieved the highest sales and exceeded its target.
- Furniture sales were below the target, indicating a need for improved sales strategies.
- Office Supplies performed close to the target, showing stable demand.
- Categories exceeding targets contributed significantly to overall revenue growth.

- **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.



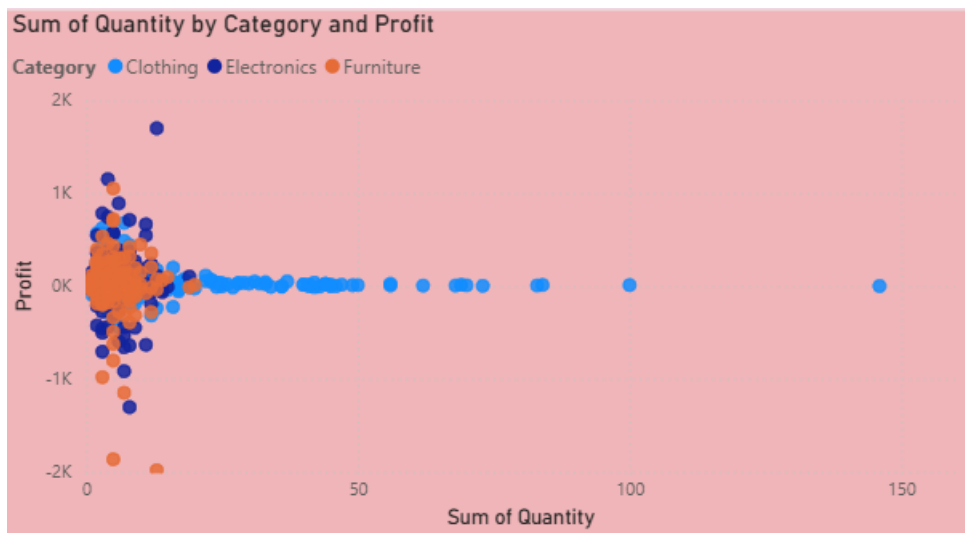
- High-margin products contribute more to profitability than high-volume products.
- Focusing on top-margin sub-categories can improve overall business profits.

- **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.



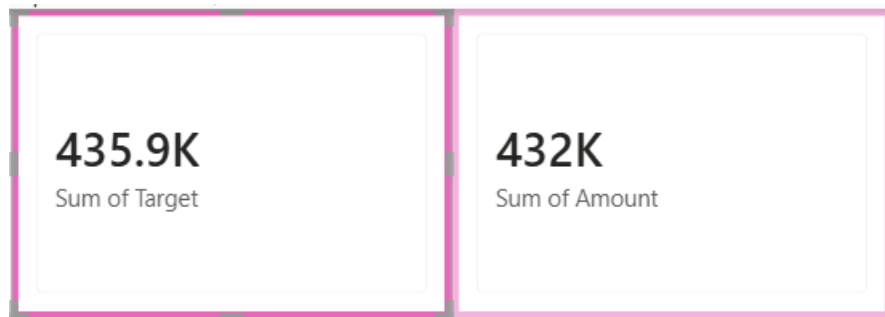
- Peak sales occurred during the holiday/festival season in the last quarter.
- Certain months experienced sales dips, indicating seasonal fluctuations.
- Monthly trends help forecast future demand and inventory requirements.

• **Comparison of Profit and Quantity by Sub-Category: Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.**



- Some sub-categories achieved high profits with relatively low quantities sold.
- High quantity sales do not always result in high profits.
- Copiers and Phones generated strong profits despite moderate sales volumes.
- Tables and Bookcases sold in higher quantities but yielded lower profits.

• **Comparison of Total Sales Amount and Target: Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.**



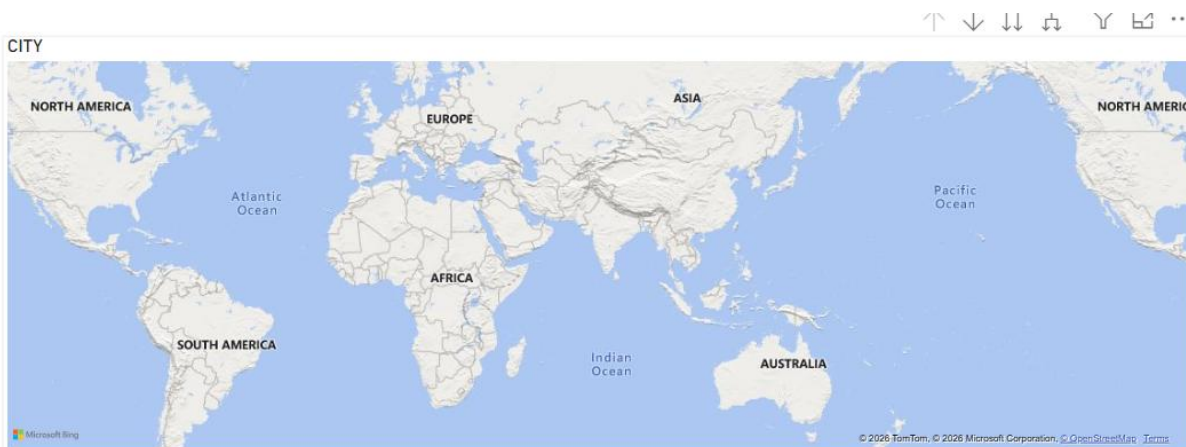
Clothing
12K MinTarget
Electronics
9K MinTarget
Furniture
10K MinTarget

- If actual sales exceed the target, the company is performing above expectations.
 - The gap between actual sales and target highlights opportunities for improvement.
 - Consumer segment has the highest minimum target requirement.
 - Home Office segment shows the lowest target values.
 - Segment-wise targets help allocate resources effectively.
- **Sales Performance Matrix: Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.**

Month Name	April			August			December			Fel
Category	Total Sales	Target Sales	Variance %	Total Sales	Target Sales	Variance %	Total Sales	Target Sales	Variance %	Tot
Clothing										
Electronics										
Furniture										
Total		31400	-1.00		33900	-1.00		36400	-1.00	

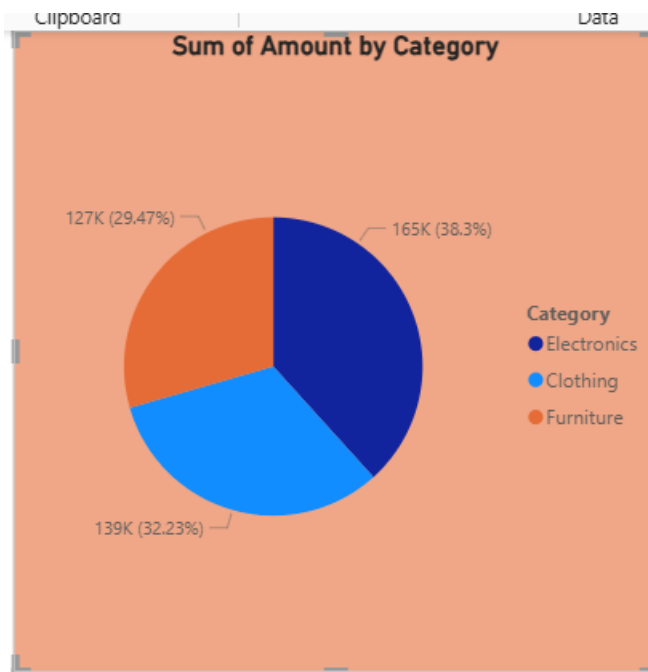
- Technology consistently outperformed targets across multiple months.
- Furniture underperformed in several months, requiring corrective actions.
- Seasonal trends are visible across categories.
- The matrix helps identify categories and months needing attention.

• **Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.**

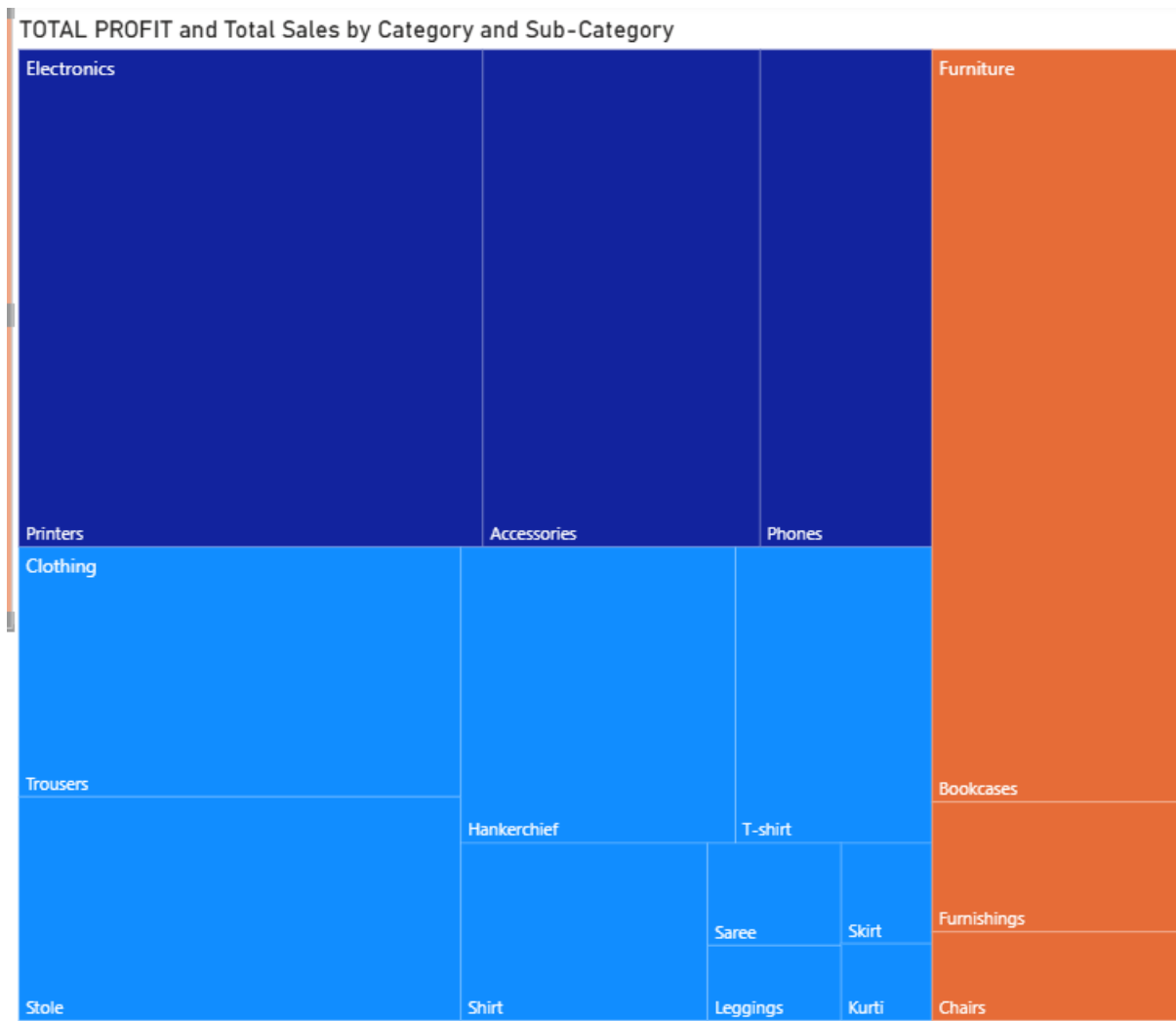


- Major cities contribute the highest share of total sales.
- Sales are concentrated in urban regions.
- Certain cities show untapped market potential due to lower sales volumes.
- Regional analysis supports expansion and marketing decisions.
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• **Sales Distribution by Sub-Category: Represent the sales distribution across Different**

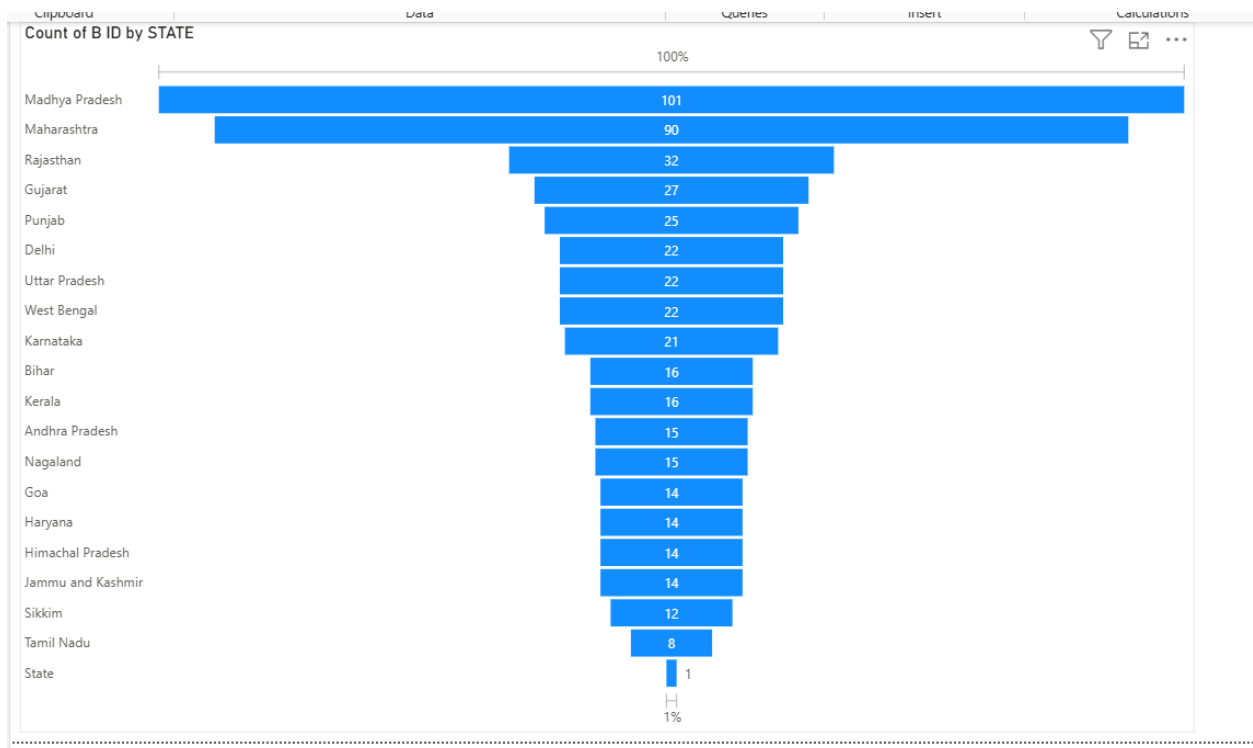


- **sub-categories using a treemap.**



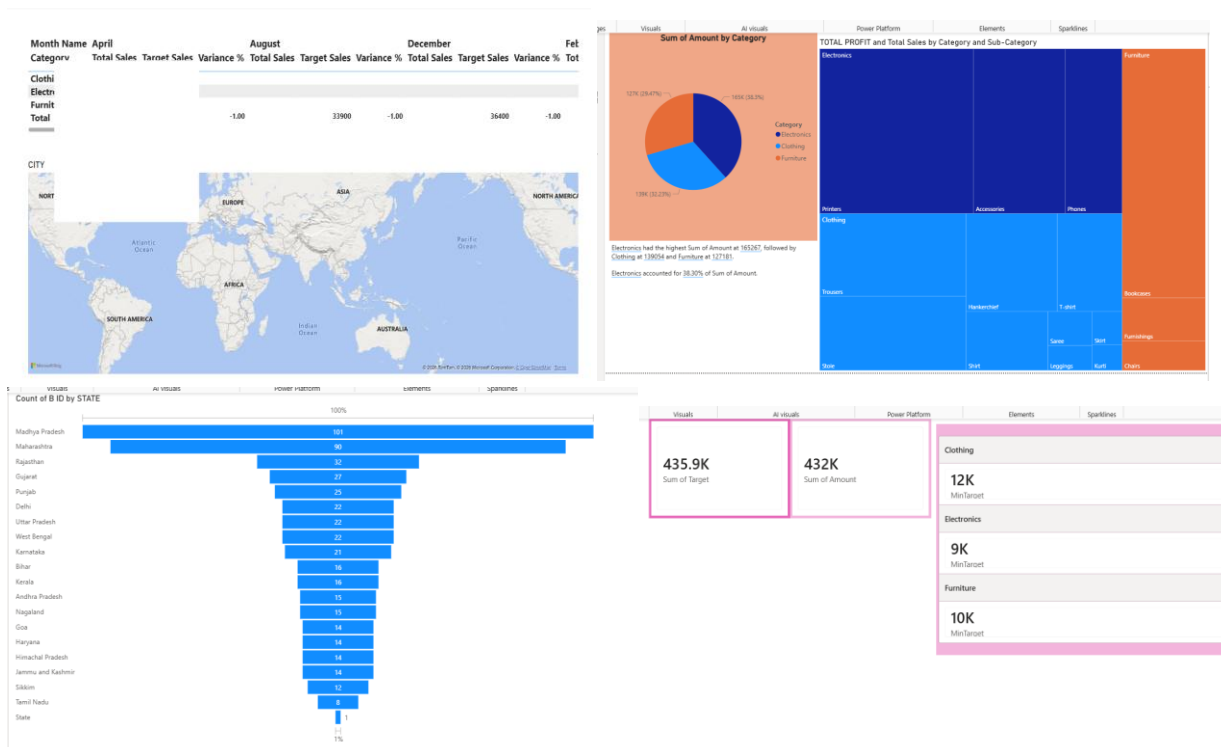
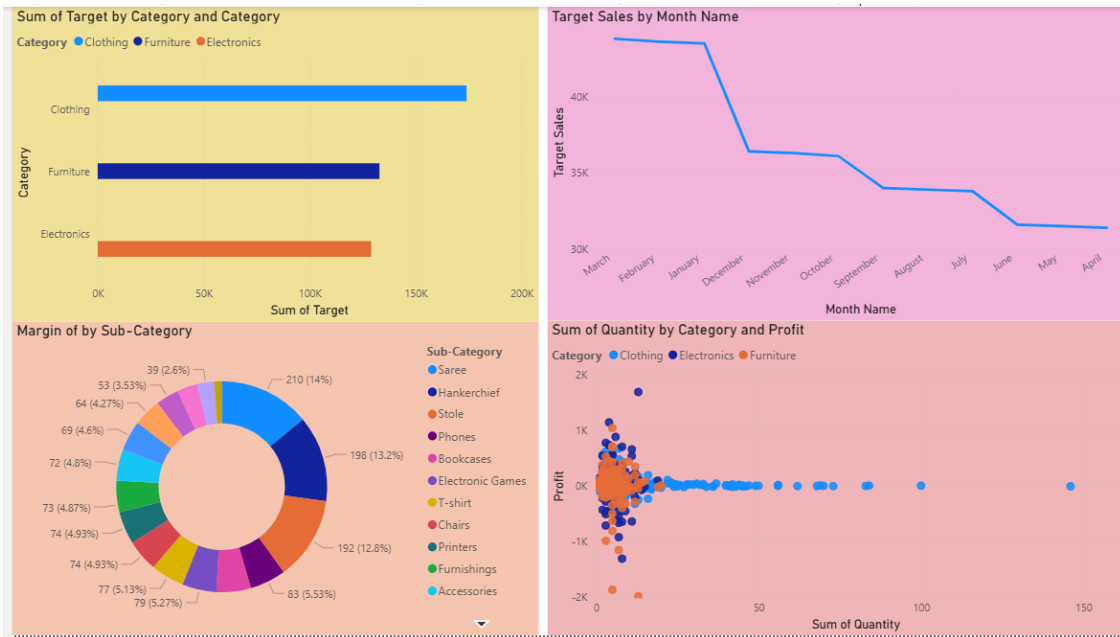
- Phones, Chairs, and Storage contribute the largest portion of sales.
- Several sub-categories contribute only a small percentage of total revenue.
- Sales are concentrated among a few high-performing product groups.
- Treemap visualization helps identify revenue-driving products.

• **Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.**



- California recorded the highest number of orders.
- Order volume gradually decreases across other states.
- A small number of states contribute a majority of total orders.
- States with low order counts represent potential growth opportunities.

DASHBOARD



INSIGHTS

- The sales analysis shows that Technology is the best-performing category, achieving or exceeding its sales targets, while Furniture requires improvement.

- Copiers and Phones generate the highest profit margins, contributing significantly to overall profitability.
- Monthly sales display an upward trend with higher sales during peak seasons.
- Geographic analysis reveals that major cities and states contribute the most sales and orders, while some regions offer growth opportunities. The sales distribution indicates that a few key sub-categories drive most of the revenue.
- Overall, the business demonstrates strong sales performance, but focusing on underperforming categories and regions can further improve growth and profitability.